Methods and Determination of Quadrupole β_2 Deformation Parameters Across Experimental Techniques

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Abstract

Contents

Abstract **1**

1 Introduction and Theory **3**

1.1 Quadrupole Deformation Overview 3

1.2 Intrinsic Quadrupole Moment 5

1.3 Reduced Transition Probability $B(E2)$ 5

1.4 Key Relations for Extracting β_2 5

2 Experimental Methods **5**

2.1 Principle of Coulomb Excitation 5

2.2 Inverse Kinematics 5

2.3 Target Considerations 5

2.4 Experimental Observables 5

2.5 Detector Systems 5

3 Methodology for Extracting $B(E2)$ and β_2 **5**

3.1 Gamma-Ray Identification 5

3.2 Efficiency Corrections 5
3.3 Angular Analysis 5
3.4 Normalization Procedures 5
3.5 GOSIA Analysis 5
3.6 Extraction of E2 Matrix Elements 5
3.7 Calculation of $B(E2)$ 5
3.8 Calculation of β_2 5

4 Selected Results from Literature **5**
4.1 ^{30}Mg 5
4.2 ^{32}Mg 5
4.3 ^{74}Zn 5
4.4 ^{76}Zn 5

5 Conclusion **5**

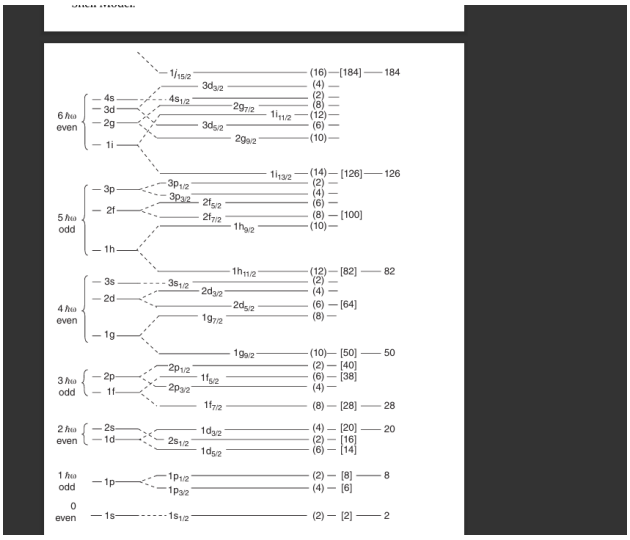


Figure 1: Example of a shell model spectroscopic chart based on how nucleons fill orbitals (harmonic oscillator + Woods-Saxon + spin/orbit). Image taken from [1].

1 Introduction and Theory

This paper is a compact discussion about how the quadrupole deformation parameters β_2 are measured from reduced transition probabilities $B(E2)$ for a 0^+ to 2^+ transition. Specifically, the Coulomb excitation methodology is the main driver for this measurement, along with simulation tools such as GOSIA. This chapter provides a brief discussion of the theoretical background for defining β_2 and $B(E2)$, including their relationship and how to measure them. The following theoretical background entails starting with the shell model breakdown and how it leads to cases of nonspherical considerations, what the quadrupole formation is, the intrinsic quadrupole moment, and how this is all related to the transition probability $B(E2)$ and deformation parameters β_2.

1.1 Quadrupole Deformation Overview

The story of nuclear structure always starts with the shell model. Essentially, the shell model is an attempt to explain nuclear structure (how nucleons fill orbitals), where protons and neutrons for isotopes can have specific orbital filling schemes (see 1) that explain certain parity, total angular momentum, and other aspects such as magnetic moments, quadrupole moments and Most importantly, the shell model explains how certain isotopes can be more stable than others by having closed shells, or what we call ’”magic numbers”. As shown in Figure 1, the shell model is governed by a Hamiltonian combining the harmonic oscillator, Woods-Saxon, and spin-orbit coupling as a mean field potential. This is important to note because the combination of all three is what gave correct approximations for achieving magic numbers [1].

1.2 Intrinsic Quadrupole Moment

1.3 Reduced Transition Probability $B(E2)$

1.4 Key Relations for Extracting β_2

2 Experimental Methods

2.1 Principle of Coulomb Excitation

2.2 Inverse Kinematics

2.3 Target Considerations

2.4 Experimental Observables

2.5 Detector Systems

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